Homological Algebra

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October 21, 2026

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Chapter 1

Some notes from 2026 - class

1.1 Derived functors

- (Abelian Categories)TODO:
- (Ajoints) Take a functor $F : \mathcal{C} \to \underline{\mathbf{Set}}$, a object $A \in \mathcal{C}$ and an element $a \in FA$. Define the map

$$a_B : \mathrm{Hom}_{\mathcal{C}}(A, B) \to F(B) : k \mapsto F(k)a. \tag{1.1}$$

Then $\eta(a) : \mathrm{Hom}_{\mathcal{C}}(A, -) \Rightarrow F$ is a natural transformation which sends $B \mapsto a_B$. Moreover, we have a bijection

$$A \ni a \mapsto \eta(a) \in \{\eta : \mathrm{Hom}_{\mathcal{C}}(A, -) \Rightarrow F\}. \tag{1.2}$$

The inverse of this map is given by

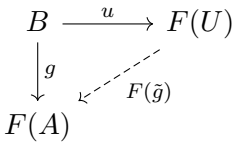
$$\{\eta : \mathrm{Hom}_{\mathcal{C}}(A, -) \Rightarrow F\} \to A : \eta_A \mapsto \eta_A(\mathbb{1}_A). \tag{1.3}$$

In fact, if $a = \eta_A(\mathbb{1}_A)$, then

$$\begin{aligned} a_B(k) &= F(k)a \\ &= F(k)(\eta_A(\mathbb{1}_A)) \\ &= \eta_B \mathrm{Hom}_{\mathcal{C}}(A, k)(\mathbb{1}_A) \quad \text{(because } \eta_A \text{ is a nat. transf.)} \\ &= \eta_B(k \circ \mathbb{1}_A) \\ &= \eta_B(k). \end{aligned} \tag{1.4}$$

Therefore $a_B = \eta_B$. In the context of the Yoneda's lemma, we say that the pair (A, a), $a = \eta_A(\mathbb{1}_A)$ represents the natural transformation $\eta_A : \mathrm{Hom}_{\mathcal{C}}(A, -) \Rightarrow F$.

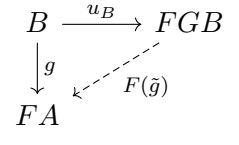
Definition 1.1.1. Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. Then we say that the pair (U, u) is a universal from B to F if for any $g \in \mathrm{Hom}_{\mathcal{D}}(B, FA)$ there exists a unique $\tilde{g} \in \mathrm{Hom}_{\mathcal{C}}(U, A)$ such that



Example 1.1.2. Let $\mathfrak{g}$, then the universal enveloping algebra $\mathcal{U}(\mathfrak{g})$ is a universal from $\mathfrak{g}$ to the functor $F : \underline{\mathbf{Ass}} \to \underline{\mathbf{Lie}}$ that takes an associative algebra A and returns the Lie algebra $(A, [\cdot, \cdot])$ where $[a, b] = ab - ba$.

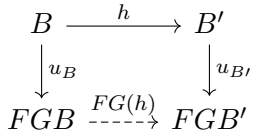
Proposition 1.1.3. Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. Suppose that for each $B \in \mathcal{D}$, there exists a universal (GB, u_B) from B to F. We claim that G is a functor from $\mathcal{D} \to \mathcal{C}$ and there exist a natural isomorphism $\mathrm{Hom}_{\mathcal{C}}(GB, -) \Rightarrow \mathrm{Hom}_{\mathcal{D}}(B, -) \circ F$. In this condition we say that G is left adjoint to F.

Proof: Let $g \in \mathrm{Hom}_{\mathcal{D}}(B, FA)$, then by definition of the universal, there exists a (unique) $\tilde{g} \in \mathrm{Hom}_{\mathcal{C}}(GB, A)$ such that the diagram commutes.



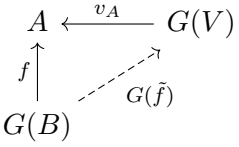
So the map $\eta_{B,A} : \mathrm{Hom}_{\mathcal{C}}(GB, A) \to \mathrm{Hom}_{\mathcal{D}}(B, FA) : f \mapsto F(f)u_B$ is bijective (in fact, the universal condition is equivalent to bijection of $\eta_{B,A}$). Therefore $A \mapsto \eta_{B,A}$ gives a natural isomorphism $\mathrm{Hom}_{\mathcal{C}}(GB, -) \Rightarrow \mathrm{Hom}_{\mathcal{D}}(B, -) \circ F$.

Now, it remains to prove that G is a functor. So if $B, B' \in \mathcal{D}$ and $h \in \mathrm{Hom}_{\mathcal{D}}(B, B')$, we need to construct a morphism $G(h) \in \mathrm{Hom}_{\mathcal{C}}(GB, GB')$, we define it to be the unique morphism making the diagram below commute.



Since (GB, u_B) are universals, it's easy to see that $G(h \circ h') = G(h) \circ G(h')$ and $G(\mathbb{1}_B) = \mathbb{1}_{GB}$. □

Remark 1.1.4. The concept of universal has a dual: Let $G : \mathcal{D} \to \mathcal{C}$ be a functor. We say that (V, v) is a universal from G to $A \in \mathcal{C}$ if for any morphism $f \in \mathrm{Hom}_{\mathcal{C}}(GB, A)$, there exist a (unique) morphism $\tilde{f} \in \mathrm{Hom}_{\mathcal{D}}(B, V)$ such that the diagram below commutes.



Proposition 1.1.5. Let G be as in the previous proposition. For every $A \in \mathcal{C}$, define $v_A : GFA \to A$ as the unique morphism making the diagram below commutes.

$$\begin{array}{ccc} FA & \xrightarrow{u_{FA}} & FGFA \\ \downarrow{\scriptstyle \mathbb{1}_{FA}} & \swarrow_{F(v_A)} & \\ FA & & \end{array}$$

Then for each $A \in \mathcal{C}$, the pair (FA, v_A) is a universal from G to A. In this condition there exist a natural isomorphism $\mathrm{Hom}_{\mathcal{D}}(-, FA) \Rightarrow \mathrm{Hom}_{\mathcal{C}}(-, A) \circ G$. So we say that F is right adjoint to G.

Proof: The claim that (FA, v_A) is a universal from G to A is equivalent to the claim that $\xi_{B,A} : \mathrm{Hom}_{\mathcal{D}}(B, FA) \to \mathrm{Hom}_{\mathcal{C}}(GB, A) : f \mapsto v_A G(f)$ is a bijection. So we prove this instead. We have

$$\begin{aligned} \eta_{B,A}\xi_{B,A}(f) &= F(v_A G(f))u_B \\ &= F(v_A)FG(f)u_B \quad (F \text{ is a functor}) \\ &= F(v_A)u_{FA}f \quad (\text{definition of } G(f), \text{ see diagram (1.1)}) \\ &= \mathbb{1}_{FA}f \\ &= f. \end{aligned} \tag{1.5}$$

So we see that $\xi_{B,A} = \eta_{B,A}^{-1}$, which is a bijection. □

- (Homotopy) We define a equivalence relation on $\mathrm{Hom}(C_\bullet, D_\bullet)$ as follows: Let $f, g : (C_\bullet, d) \to (D_\bullet, d)$ be two morphims of complexes, then we say that $f \sim g$ if there exist $h_n : C_n \to D_{n+1}$, $\forall n$ such that the diagram below commutes.

$$\begin{array}{ccccccccc} C_{n-1} & \xleftarrow[d_n]{} & C_n & \xleftarrow[d_{n+1}]{} & C_{n+1} & \longleftarrow & \cdots \\ \downarrow{\scriptstyle h_{n-1}} & \searrow & \downarrow{\scriptstyle h_n} & \searrow & \downarrow{\scriptstyle h_{n+1}} & \searrow & \\ D_{n-1} & \xleftarrow[d_n]{} & D_n & \xleftarrow[d_{n+1}]{} & D_{n+1} & \longleftarrow & \cdots \end{array}$$

We write $h \circ d + d \circ h = g - f$. Observe that d decreases the index and h increases the index. So $(h \circ d)_n = h_{n-1} \circ d_n$ and $(d \circ h)_n = d_{n+1} \circ h_n$.

Definition 1.1.6. We say that $f_\bullet : C_\bullet \to D_\bullet$ is a Homotopy equivalence if there exists $g_\bullet : D_\bullet \to C_\bullet$ such that $g_\bullet \circ f_\bullet \sim \mathrm{Id}_{C_\bullet}$ and $f_\bullet \circ g_\bullet \sim \mathrm{Id}_{D_\bullet}$.

Notation 1.1.7. Normally we will avoid to write $(C_\bullet, d)$ for complexes and adote the minimalistic notation by just writing "C" instead. Same with morphism of complexes.

Proposition 1.1.8. Let $f, g : C \to D, f \sim g$, then $H_n(f) = H_n(g), \forall n$.

Proof: Remember that we define $H_n(f) : H_n(C) \to H_n(D) : [x] \mapsto [f(x)]$. But since $g - f = hd + dh$, we get that $g(x) - f(x) = dh(x), x \in \mathrm{Ker}(d : C_n \to C_{n-1})$ and $[f(x)] = [g(x)]$. □

Corollary 1.1.9. If $f : C \to D$ is Homotopy equivalence then $H_n(C) \cong H_n(D)$.

Proof: By hypothesis, there exists $g : D \to C$ such that $f \circ g \sim \mathrm{Id}_D, g \circ f \sim \mathrm{Id}_C$. Therefore $H_n(f) \circ H_n(g) = H_n(\mathrm{Id}_D) = \mathrm{Id}_{H_n(D)}$ and etc. □

- (Injective resolution) Here $\mathcal{C}$ denotes an abelian category if not stated otherwise.

  Definition 1.1.10. We say that $I \in \mathcal{C}$ is injective if $\mathrm{Hom}_{\mathcal{C}}(-, I) : \mathcal{C} \to \underline{\mathrm{Ab}}$ is an an exact functor. This is equivalent to say that whenever we have a monomorphism $i : A \to B$ and a morphism $f : A \to I$, then there exists $\widetilde{f} : B \to I$.

$$\begin{array}{ccc} A & \xrightarrow{i} & B \\ {\scriptstyle f}\downarrow & \swarrow_{\exists \widetilde{f}} & \\ I & & \end{array}$$

  Definition 1.1.11. Let $A \in \mathcal{C}$, then an injective resolution for A is a complex

$$I^\bullet = (0 \to I^0 \to I^1 \to I^2 \to \cdots) \tag{1.6}$$

  such that

  (1) I^n is injective;
  (2) The augmented complex $\widetilde{I^\bullet} = (0 \to A \to I^0 \to I^1 \to \cdots)$ is acyclic.

  Definition 1.1.12. We say that $\mathcal{C}$ has enough injectives if for any $A \in \mathcal{C}$ there exists an injective object $I \in \mathcal{C}$ and a monomorphism $f : A \to I$.

  Proposition 1.1.13. If $\mathcal{C}$ has enough injectives, then any $A \in \mathcal{C}$ has a injective resolution.

  Proof: Draw the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{\epsilon} & I^0 & \dashrightarrow & & \dashrightarrow & I^1 \cdots \\ & & & & & \searrow & & \nearrow & \\ & & & & & & I^0/\epsilon(A) = Q^0 & & \end{array}$$

  and repeat

$$\begin{array}{ccccc} I^1 & \dashrightarrow & & \dashrightarrow & I^2 \cdots \\ & \searrow & & \nearrow & \\ & & I^1/Q^0 = Q^1 & & \end{array}$$

etc... □

- (Derived functors) Assume that $\mathcal{A}$ has enought injectives. If $F : \mathcal{A} \to \mathcal{B}$ is a left exact functor we define
$$R^n F(A) = H^n(F(I^\bullet)) \tag{1.7}$$
where $I^\bullet$ is some injective resolution of A.

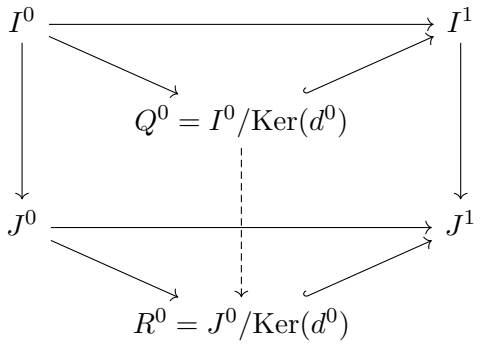
Remark 1.1.14. We have
$$\begin{aligned} R^0F(A) &= H^0(F(I^\bullet)) \\ &= H^0(0 \to F(I^0) \to F(I^1) \to \cdots) \\ &= \mathrm{Ker}(F(I^0) \to F(I^1))/\mathrm{Im}(0 \to F(I^0)) \\ &= F(A) \end{aligned} \tag{1.8}$$
Here we used that F is left exact, so $0 \to F(A) \to F(I^0) \to F(I^1)$ is exact.

Notation 1.1.15. From now on, we will adopt the following convention: We will draw arrows aways from left to right. So, index decreases from left to right in homology and increases from left to right in cohomology.
$$\begin{aligned} &\text{(Homology)} \quad \cdots \to C_2 \to C_1 \to C_0 \to 0 \\ &\text{(Cohomology)} \quad 0 \to C^0 \to C^1 \to C^2 \to \cdots \end{aligned} \tag{1.9}$$

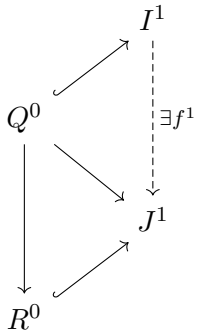
Lemma 1.1.16. Let $f : A \to B$ be a morphism and take injective resolutions $A \to I^\bullet, B \to J^\bullet$. Then there exists lifting $f^\bullet : I^\bullet \to J^\bullet$,i.e., $\{f^n\}_{n\geq 0}$ such that
$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{\eta} & I^0 & \xrightarrow{d} & I^1 & \xrightarrow{d} & \cdots \\ & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f^0} & & \downarrow{\scriptstyle f^1} & & \\ 0 & \longrightarrow & B & \xrightarrow{\eta} & J^0 & \xrightarrow{d} & J^1 & \xrightarrow{d} & \cdots \end{array}$$

Proof: First we define f^0 by the diagram below.
$$\begin{array}{ccc} A & \xrightarrow{\eta} & I^0 \\ & \searrow^{\eta\circ f} & \downarrow{\scriptstyle \exists f^0} \\ & & J^0 \end{array}$$

For the second block we construct the diagram



The dashed morphism is defined by $Q^0 \to R^0 : [x] \mapsto [f^0(x)]$. It's well defined, because if $x - x' \in \mathrm{Ker}(d^0) = \mathrm{Im}(\eta)$ implies that $x - x' = \eta(a)$. So $f^0(x) - f^0(x') = f^0\eta(a) = \eta f(a)$ and $d\eta = 0$. Now we define f^1 from the diagram

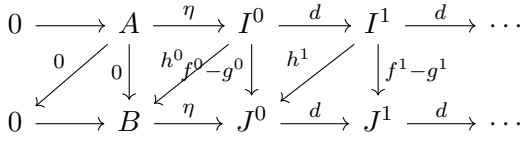


and etc... □

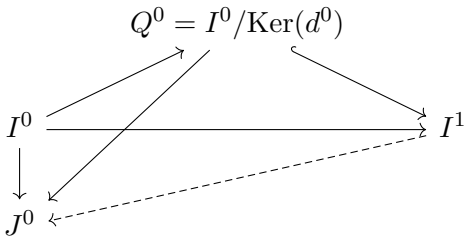
Remark 1.1.17. It looks like we only used the fact that $I^\bullet$ is acyclic and $J^\bullet$ is injective.

Lemma 1.1.18. Any two liftings are homotopic.

Proof: We want to find h (not necesseraly a morphism of complexes, it's just a map) such that makes the diagram below commutes.



For $n = 0$, we need $h^1 \circ d^0 + d^1 \circ h^0 = g^0 - f^0$. It's clear that $h^0 = 0$ makes the first block commutes, then $h^1 \circ d^0 = g^0 - f^0$. However, we have



As before, we define $Q^0 \to J^0 : x + \mathrm{Ker}(d^0) \mapsto f^0(x) - g^0(x)$. Now it's well defined because if $x \in \mathrm{Ker}(d^0) = \mathrm{Im}(\eta)$, we get $f^0(x) - g^0(x) = (f^0 - g^0)\eta(a) = 0$. □

Remark 1.1.19. To drawn the arrows of the Homotopy map, we use the rule: the direction of the arrows goes in the same direction of the chain morphisms and to the contrary direction of the chain differentials.

Remark 1.1.20. It looks like we only used the fact that $I^\bullet$ is acyclic and $J^\bullet$ is injective.

Theorem 1.1.21. (1) $RF^n(A)$ independ of choice of resolution $I^\bullet$;

(2) $f : A \to B$ induces $R^nF(A) \to R^nF(B)$;

(3) If $0 \to A \to B \to C \to 0$ is a SES, then there exists a long exact sequence:

$$\cdots \to R^1F(A) \to R^1F(B) \to R^1F(C) \to 0 \to F(A) \to F(B) \to F(C)$$

Proof: Let $A \in \mathcal{C}$ and take two injective resolutions $I^\bullet, J^\bullet$ of A. Then the identity morphism $A \to A$ lifts to morphisms $f : I^\bullet \to J^\bullet, g : J^\bullet \to I^\bullet$. In particular $fg : I^\bullet \to I^\bullet$ is a complex morphism, so it must be homotopic to the identity morphism, i.e., $fg \sim \mathrm{Id}_{I^\bullet} \Rightarrow F(f) \circ F(g) \sim \mathrm{Id}_{F(I^\bullet)}$. In a similar way we see that $F(g) \circ F(f) \sim \mathrm{Id}_{F(J^\bullet)}$, so $H^n(F(I^\bullet)) \cong H^n(F(J^\bullet))$. The second one also follows directly from the lemmas above.

TODO: Prove (3), use Horseshoe lemma

- (Projectives)
- (Group cohomology)
- (Lie algebra cohomology)

Chapter 2

Exercises

2.1 List 1

Problem 1. State the definition of monomorphism, and prove that an injective function of sets is a monomorphism in Set.

Solution: A monomorphism $f \in \mathrm{Hom}_{\mathcal{C}}(B, C)$ is a morphism that is left cancelative in the sense that for any object A and morphisms $g, h \in \mathrm{Hom}_{\mathcal{C}}(A, B)$, we have:

$$fg = fh \Rightarrow g = h. \tag{2.1}$$

Now, if $\mathcal{C} = \mathbf{Set}$, the equality $fg = fh$ implies $f(g(a)) = f(h(a)),\ \forall a \in A$ and if f injective, by definition, we have $g(a) = h(a),\ \forall a \in A$, i.e., $g = h$.

Problem 2. Prove that $h : \mathbb{Z} \hookrightarrow \mathbb{Q}$ is an epimorphism in Ring.

Solution: Let $f, g : \mathbb{Q} \to R$ be two ring morphisms such that $fh = gh$, i.e., $f(n) = g(n),\ \forall n \in \mathbb{Z}$. Here, we assume that a morphism in Ring send unity to unity, so $f(1) = g(1) = 1_R$. Furthermore, since $\mathbb{Q}$ is a field, $f : \mathbb{Q} \hookrightarrow R$ and $f(\mathbb{Q}^{\times}) \subset R^{\times}$. In particular $m1_R \in R^{\times}$. Now we have

$$m1_R(f(\frac{1}{m}) - g(\frac{1}{m})) = 0 \Rightarrow f(\frac{1}{m}) = g(\frac{1}{m}). \tag{2.2}$$

This proves that $f = g$.

Problem 3. Write the proof that any two terminal objects in a category $\mathcal{C}$ are isomorphic to each other.

Solution: Let T, T' be two terminal objects in $\mathcal{C}$. Let $f \in \mathrm{Hom}_{\mathcal{C}}(T, T')$ and $T' \in \mathrm{Hom}_{\mathcal{C}}(T', T)$, then $f \circ g \in \mathrm{Hom}_{\mathcal{C}}(T, T) = \{\mathbb{1}_T\}$. Therefore $fg = \mathbb{1}_T$ and in a similar way we see that $gf = \mathbb{1}_{T'}$. This proves that f is an isomorphism.

Problem 4. State an "explicit" definition of the free product $G * H$ of groups G and H. Prove that the free product is a coproduct in Grp.

Solution: Let $A = G \coprod H$, we define

$$G * H = \coprod_{n \geq 1} A^n \Big/ \sim \tag{2.3}$$

where we the equivalence relation is defined by

$$\begin{aligned} &(a_1, \ldots, a_i, a_{i+1}, \ldots, a_n) \sim (a_1, \ldots, a_i a_{i+1}, \ldots, a_n) \text{ if } a_1, a_2 \in G \text{ or } a_1, a_2 \in H \\ &(a_1, \ldots, a_{i-1}, a_i, a_{i+1} \ldots, a_n) \sim (a_1, \ldots, a_{i-1}, a_{i+1} \ldots, a_n) \text{ if } a_i = 1_G \text{ or } a_i = 1_H \end{aligned} \tag{2.4}$$

The multiplication is defined by

$$[(a_1, \ldots, a_n)] * [(b_1, \ldots, b_m)] = [(a_1, \ldots, a_n, b_1, \ldots, b_m)]. \tag{2.5}$$

By the equivalence relations defined above, the multiplication rule is clearly well defined.

We see that $(G * H, *)$ is a group. The associativity and existence of unity is clear and the inverse operation is defined by

$$[(a_1, a_2, \ldots, a_n)]^{-1} = [(a_n^{-1}, \ldots, a_2^{-1}, a_1^{-1})]. \tag{2.6}$$

Now we define morphims

$$i_1 : G \to G * H : g \mapsto [(g)], \; i_2 : H \to G * H : h \mapsto [(h)]. \tag{2.7}$$

We claim that $(G * H, i_1, i_2)$ is a coproduct of G and H. Suppose that we have morphisms $f_1 : G \to K, f_2 : H \to K$, then we define a morphism

$$f : G * H \to K : [(a_1, \ldots, a_n)] \mapsto f_{j_1}(a_1) \cdots f_{j_n}(a_n) \tag{2.8}$$

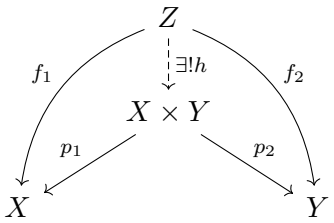
where we let $j_k = 1$ (resp. 2) if $a_k \in G$ (resp. H). Furthermore, we see that $f i_j = f_j, \; j = 1, 2$. For uniqueness, we observe that if $h : G * H \to K$ is a morphism, such that $h i_j = f_j, \; j = 1, 2$, then $h([(g)]) = f_1(g)$ and $h([(h)]) = f_2(h)$. So

$$h([(a_1, \ldots, a_n)]) = h([(a_1)] * \cdots * [(a_n)]) = h([a_1]) \cdots h([(a_n)]) \tag{2.9}$$

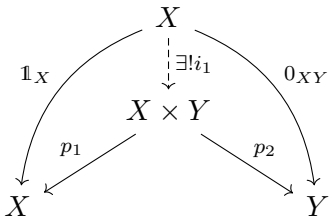
and we see that $h([(a_k)]) = f_{j_k}(a_k)$ as defined above, therefore $h = f$.

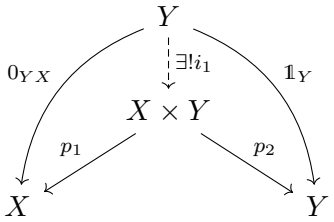
Problem 5. Let $\mathcal{C}$ be a pre-additive category and $X, Y \in \mathcal{C}$. Show that if a product $X \times Y$ of X and Y exists, then $X \times Y$ is also a coproduct of X and Y .

Solution: A product $X \times Y$ comes with projections $p_1 : X \times Y \to X$ and $p_2 : X \times Y \to Y$ such that the following universal property is satisfied.



So we define $i_1 : X \to X \times Y$ and $i_2 : Y \to X \times Y$ using this universal property:





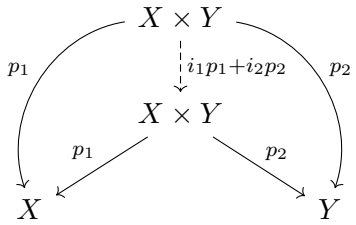
So we have the following properties

$$\begin{cases} \mathbb{1}_X = p_1 i_1,\ p_2 i_1 = 0_{XY} \\ \mathbb{1}_Y = p_2 i_2,\ p_1 i_2 = 0_{YX} \end{cases} \tag{2.10}$$

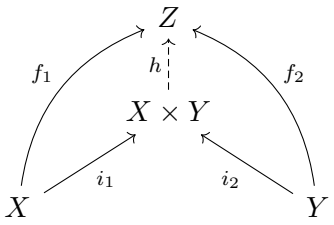
Now, we claim that

$$i_1 p_1 + i_2 p_2 = \mathbb{1}_{X \times Y} \quad (*) \tag{2.11}$$

which can be seen by the diagram below and the universal property of the product.



Now, suppose that we have morphisms $f_1 : X \to Z, f_2 : Y \to Z$. Then if we define $h = f_1 p_1 + f_2 p_2$, the diagram below commutes.

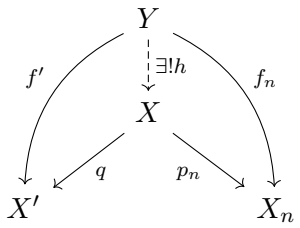


Furthermore, relations (2.10) and equation (*) guarantees that this is the unique expression possible for h that makes the above diagram commutes.

Problem 6. Show that existence of binary products implies existence of all finite n-ary products for $n \geq 2$.

Solution: By a finite n-ary product ($n \geq 2$), we mean an object $X = \prod_{i=1}^{n} X_i$ with morphims $p_i : X \to X_i$ satisfying the following universal property. Take a colection of morphims $f_i : Y \to X_i$, $(i = 1, \ldots, n)$, then there exist an unique morphism $f : Y \to X$ such that $p_i f = f_i$, $\forall i$. By induction, assume that there exist all finite $(n-1)-$ary products and let $X' = \prod_{i=1}^{n-1} X_i$. We want to show that $X = X' \times X_n$ satisfies the universal property for a $n-$ary product. The projections are obtained as follows: we have the projection $p_n : X \to X_n$ and for $i < n$ we compose the projection $q : X \to X'$ with the ith projection $p'_i : X' \to X_i$ to get $p_i : X \to X_i$.

Now assume we have morphisms $f_i : Y \to X$, $(i = 1, \ldots, n)$. Then, by the universal property of the finite $(n-1)-$ary product, there exists a (unique) morphism $f' : Y \to X'$ such that $p'_i f' = f_i$, $i < n$. Now we have the diagram:



Now we observe that

$$p_i h = (p'_i q)h = p'_i(qh) = p'_i f' = f_i, \; i < n. \tag{2.12}$$

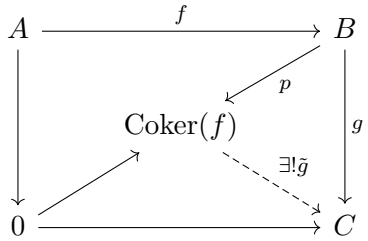
This show existence of h. Now, for uniqueness, suppose that h' is such that $p_i h' = f_i$, $\forall i$. Then, for $i < n$ we have $p'_i(h'q) = f_i$. So by the universal property of the finite $(n-1)-$ary product X', we must have $h'q = f'$. Then, by applying the universal property of the product $X = X' \times X_n$, we must have $h' = h$.

Problem 7. Consider a composition

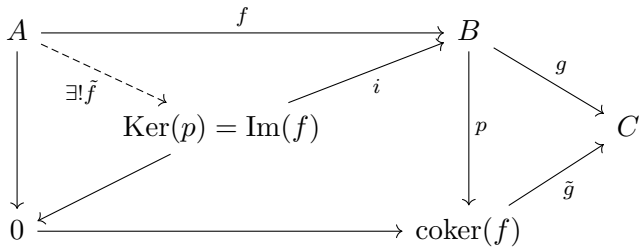
$$A \longrightarrow B \longrightarrow C$$

in an abelian category, in which $g \circ f = 0$. Using only universal properties, show that there is a canonical morphism from $\mathrm{im}(f)$ to $\ker(g)$. [Recall $\mathrm{im}(f) = \ker(\mathrm{coker}(f))$ by definition.]

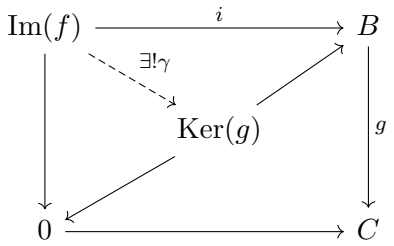
Solution: From the universal property of the cokernel we have the diagram.



Now, by the universal property of the kernel, we have:



Finally, by applying the universal property of the kernel again, we have:



as desired.

Problem 8. Let $G : \mathbf{Ab} \to \mathbf{Set}$ be the forgetful functor, and let $F : \mathbf{Set} \to \mathbf{Ab}$ be the functor of forming a free abelian group. Show that F is left adjoint to G.

Solution: We have the map

$$\mathrm{Hom}_{\mathbf{Ab}}(F(X), Y) \to \mathrm{Hom}_{\mathbf{Set}}(X, G(Y)) : f \mapsto f|_X \tag{2.13}$$

Here, we let $F(X) = \{f : X \to \mathbb{Z} : f(x) \neq 0 \text{ for a finite number of } x \in X\}$ and denote $\hat{x} \in F(X)$ for the map defined by $\hat{x}(y) = \delta_{xy}$. So we see that the set theoretic map $\iota : X \to F(X) : x \to \hat{x}$ is an injection and this justifies the notation $f|_X$ used in $(*)$. Moreover, show that this map is an isomorphism is equivalent to show that we have the following universal property: For any set map of sets $f : X \to G(Y)$, there exists a unique morphism $\hat{f} : F(X) \to Y$ such that $\hat{f}\iota = f$. So we have no choice but define $\hat{f}(\hat{x}) = f(x)$ and extend to $F(X)$ by $\mathbb{Z}-$linearity, this defines $\hat{f}$ uniquely.

Problem 9. Compute $(\mathbb{Z}/5) \otimes_{\mathbb{Z}} (\mathbb{Z}/6)$ and $(\mathbb{Z}/p^2) \otimes_{\mathbb{Z}} (\mathbb{Z}/p)$ where p is a prime.

Solution: First we claim that $(\mathbb{Z}/5) \otimes_{\mathbb{Z}} (\mathbb{Z}/6) = (0)$. In fact, if $a \otimes b \in (\mathbb{Z}/5) \otimes_{\mathbb{Z}} (\mathbb{Z}/6)$, then since $6^2 \equiv 1 \mod 5$, we have

$$a \otimes b = a \cdot 6^2 \otimes b = a \cdot 6 \otimes 6 \cdot b = a \cdot 6 \otimes 0 = 0. \tag{2.14}$$

For the second group, we claim that $(\mathbb{Z}/p^2) \otimes_{\mathbb{Z}} (\mathbb{Z}/p) \cong \mathbb{Z}/p$. To show this, we show that $\mathbb{Z}/p$ satisfies the universal property of the tensor product (here we will use the notation of problem 10). Let $\varphi_0 : \mathbb{Z}/p^2 \times \mathbb{Z}/p \to \mathbb{Z}/p : ([m], [n]) \mapsto [mn]$, it's a well defined balanced map, i.e., $\varphi_0 \in \mathrm{Bal}_{A,B}(\mathbb{Z}/p)$. In particular $\mathbb{Z}/p$ is generated by the image of φ_0. Now, let $f \in \mathrm{Bal}_{A,B}(X)$, we want to define $\tilde{f}$ in such a way the following diagram commutes.

$$\begin{array}{ccc} \mathbb{Z}/p^2 \times \mathbb{Z}/p & \xrightarrow{\varphi_0} & \mathbb{Z}/p \\ {\scriptstyle f}\big\downarrow & \swarrow_{\tilde{f}} & \\ X & & \end{array}$$

So we set $\tilde{f}([n]) = f(1, [n])$, which is well defined while makes the diagram commutes.

Problem 10. Let R be a unital ring. In order to formulate the definition of tensor product of R-modules, we introduce a definition:
definition 0.1. Let A be a right R-module and B a left R-module, and X an abelian group. A balanced map $f \in \mathrm{Bal}_{A,B}(X)$ is a function $f : A \times B \to X$ such that

$$f(a_1 + a_2, b) = f(a_1, b) + f(a_2, b) \quad \text{for all } a_1, a_2 \in A \text{ and } b \in B$$
$$f(a, b_1 + b_2) = f(a, b_1) + f(a, b_2) \quad \text{for all } a \in A \text{ and } b_1, b_2 \in B$$
$$f(ar, b) = f(a, rb) \quad \text{for all } a \in A, b \in B \text{ and } r \in R.$$

Now a tensor product of a right R-module A with a left R-module B is an abelian group $A \otimes_R B$ together with a $\varphi_0 \in \mathrm{Bal}_{A,B}(A \otimes_R B)$, such that for any balanced map f as below, there exists a unique linear map $\tilde{f}$ making the diagram commute.

$$\begin{array}{ccc} A \times B & \xrightarrow{\varphi_0} & A \otimes_R B \\ & {\scriptstyle f} \searrow \quad \swarrow {\scriptstyle \tilde{f}} & \\ & X & \end{array}$$

Verify that

$$A \otimes_R B \cong Q = \frac{A \otimes_{\mathbb{Z}} B}{\langle ar \otimes b - a \otimes rb \mid a \in A, b \in B, r \in R \rangle}.$$

Solution: To prove the asserted isomorphism, we show that the quotient of abelian groups Q satisfies the universal property of the tensor product $A \otimes_R B$. Let $f \in \mathrm{Bal}_{A,B}(X)$, then f is balanced by viewing A, B as abelian groups. By the universal property of $A \otimes_{\mathbb{Z}} B$, there exist maps $\hat{\varphi}_0$ and $\hat{f}$, such that:

$$\begin{array}{ccc} A \times B & \xrightarrow{\hat{\varphi}_0} & A \otimes_{\mathbb{Z}} B \\ & {\scriptstyle f} \searrow \quad \swarrow {\scriptstyle \hat{f}} & \\ & X & \end{array}$$

Now, since $f \in \mathrm{Bal}_{A,B}(X)$, we have

$$\mathrm{Ker}(\hat{f}) \supset \langle ar \otimes b - a \otimes rb \mid a \in A, b \in B, r \in R \rangle. \tag{2.15}$$

Then, there exists $\tilde{f}$ such that the following diagram commutes:

$$\begin{array}{ccccc} A \times B & \xrightarrow{\hat{\varphi}_0} & A \otimes_{\mathbb{Z}} B & \xrightarrow{p} & Q \\ & {\scriptstyle f} \searrow & \swarrow {\scriptstyle \hat{f}} & \swarrow {\scriptstyle \tilde{f}} & \\ & & X & & \end{array}$$

Now, by construction, the quotient Q is generated by the image of the map $\varphi_0 = p\hat{\varphi}_0$ and since $f = \tilde{f}\varphi_0$ we see that $\tilde{f}$ is uniquely determined by f. Therefore (Q, φ_0) satisfies the universal property for the tensor product $A \otimes_R B$ and $Q \cong A \otimes_R B$.

2.2 List 2

Problem 1. Let A be an abelian group, and consider the action

$$A \times A \longrightarrow A$$

given by addition in the group. Does this action make A into an A-module? Justify your answer.

Solution: For $a, m \in A$ we define $am = a + m$. One of the module axioms says $(a + a')m = am + a'm$, so we must have

$$a + a' + m = a + m + a' + m \Rightarrow m = 0. \tag{2.16}$$

So, the given action defines an $A-$module if and only if A is the trivial group. □

Problem 2. The homologies $H_n(G, A)$ of a group G with coefficients in the G-module A are the derived functors of the coinvariants functor

$$(-)_G = \mathbb{Z} \otimes_{\mathbb{Z}G} (-).$$

(where $\mathbb{Z}$ is the trivial G-module). Using a convenient projective resolution of $\mathbb{Z}$, or otherwise, compute group homology $H_n(G, A)$ where G is the cyclic group $\mathbb{Z}/n$ and A is a trivial G-module.

Solution: We use the periodic projective resolution

$$\cdots \overset{g-1}{\to} \mathbb{Z}G \overset{N}{\to} \mathbb{Z}G \overset{g-1}{\to} \mathbb{Z}G \overset{\epsilon}{\to} \mathbb{Z} \tag{2.17}$$

where $G = C_n = \langle g : g^n = 1 \rangle$, $\epsilon : g \mapsto 1$ and $N = 1 + g + \cdots + g^{n-1}$. Now we apply $(-) \otimes_{\mathbb{Z}G} A$ (which is left exact) to the above sequence to get

$$\cdots \overset{g-1}{\to} A \overset{N}{\to} A \overset{g-1}{\to} A \overset{\epsilon}{\to} A_G. \tag{2.18}$$

Here we used the general fact that if M is a R-module, then $R \otimes_R M \cong M$ as $R-$modules. Now we have

$$\begin{aligned} H_n(G, A) &= [R^n\mathbb{Z} \otimes_{\mathbb{Z}G} (-)](A) \quad \text{(definition)} \\ &= [L_n(-) \otimes_{\mathbb{Z}} A](\mathbb{Z}) \quad \text{(balancing)} \\ &= H_n(\cdots \overset{g-1}{\to} A \overset{N}{\to} A \overset{g-1}{\to} A \to 0). \end{aligned} \tag{2.19}$$

Now, if A is a trivial module, then $ga = a, \forall g \in G$. Therefore $\mathrm{Ker}(g-1) = A$ and $\mathrm{Im}(N) = nA$. So we get

$$H_m(G,A) = \begin{cases} A/nA, \ m \text{ odd}; \\ \{a \in A : na = 0\}, \ m \text{ even } > 0; \\ A_G, \ m = 0 \end{cases} \tag{2.20}$$

□

Problem 3. **The Curry adjunction**: Let $A \in \mathbf{Set}$ be a fixed set, and let $L : \mathbf{Set} \to \mathbf{Set}$ and $R : \mathbf{Set} \to \mathbf{Set}$ be the functors

$$L(X) = X \times A \qquad \text{and} \qquad R(X) = \mathrm{Hom}_{\mathbf{Set}}(A, X).$$

Prove that L is left adjoint to R, i.e., construct a natural isomorphism

$$\mathrm{curry} : \mathrm{Hom}_{\mathbf{Set}}(X \times A, Y) \longrightarrow \mathrm{Hom}_{\mathbf{Set}}\big(X, \mathrm{Hom}_{\mathbf{Set}}(A, Y)\big).$$

Solution: Let $X \in \underline{\mathbf{Set}}$, we claim that $(X \times A, u_X)$, $u_X : X \to R(X \times A) : x \mapsto [a \mapsto (x, a)]$ is a universal from X to R. Let $f \in \mathrm{Hom}_{\underline{\mathbf{Set}}}(X, R(Y))$, if we define $\tilde{f} : X \times A \to Y : (x, a) \mapsto f(x)(a)$, then the diagram below commutes

$$\begin{array}{ccc} X & \xrightarrow{u_X} & R(X \times A) \\ \downarrow{\scriptstyle f} & \swarrow_{R(\tilde{f})} & \\ R(Y) & & \end{array}$$

In fact $R(\tilde{f})u_X = \mathrm{Hom}(A, \tilde{f})u_X = \tilde{f} \circ u_X$ and $\tilde{f} \circ u_X(x)(a) = \tilde{f}(x, a) = f(x)(a)$. In particular, we see that commutativity of the diagram uniquely defines $\tilde{f}$. Then we see that for each $X \in \underline{\mathbf{Set}}$, there exist a universal (LX, u_X) from X to R. This statement is equivalent to say that the map $\eta_{X,Y} : \mathrm{Hom}_{\underline{\mathbf{Set}}}(LX, Y) \to \mathrm{Hom}_{\underline{\mathbf{Set}}}(X, RY) : g \mapsto R(g)u_X$ is an isomorphism, i.e., the assigment $Y \mapsto \eta_{X,Y}$ gives a natural isomorphism $\mathrm{Hom}_{\underline{\mathbf{Set}}}(LX, -) \Rightarrow \mathrm{Hom}_{\underline{\mathbf{Set}}}(X, R-)$. This proves that L is left adjoint to R. □

Problem 4. Prove the following lemma from class.

Lemma 0.1. If $L : \mathcal{A} \to \mathcal{B}$ is a functor with right adjoint R, and

$$\eta : \mathrm{Id} \Rightarrow R \circ L \qquad \text{and} \qquad \varepsilon : L \circ R \Rightarrow \mathrm{Id}$$

are as in class (see Corollary 3.4.9). Then

$$\varepsilon_{L(X)} \circ L(\eta_X) = \mathrm{Id}_{L(X)}\,.$$

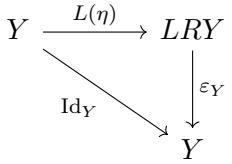
Hint: There are several equivalent definitions of adjunction. The one I have given is in terms of a natural isomorphism

$$\beta_{X,Y} : \mathrm{Hom}(L(X), Y) \longrightarrow \mathrm{Hom}(X, R(Y)).$$

There are other definitions, in particular one that is in terms of two natural transformations

$$\eta : \mathrm{Id}_{\mathcal{A}} \Rightarrow R \circ L, \qquad \varepsilon : L \circ R \Rightarrow \mathrm{Id}_{\mathcal{B}}\,.$$

One of the axioms in this second definition is



The identity to be proved follows fairly easily from this axiom. So the exercise, really, is to understand this second definition of adjoint functors, and why it follows from the first.

Solution: In the context of the Yoneda's lemma, we have a bijection

$$A \to \{\eta : \mathrm{Hom}_{\mathcal{C}}(A, -) \Rightarrow F\} : a \mapsto [a_B : \mathrm{Hom}_{\mathcal{C}}(A, B) \to F(B) : k \mapsto F(k)a]. \tag{2.21}$$

Its inverse is given by $\eta_A \mapsto \eta_A(\mathbb{1}_A)$. In this context we say that (A, a), $a = \eta_A(\mathbb{1}_A)$ represents η_A. Now, coming back to the problem, we have a natural isomorphism $\eta_{LX} : \mathrm{Hom}_{\mathcal{A}}(LX, -) \Rightarrow \mathrm{Hom}_{\mathcal{B}}(X, R-)$ given by the assigment $Y \mapsto \eta_{X,Y}$. So, by the Yoneda's lemma, we see that the pair $(LX, u_X), u_X = \beta_{X,LX}(\mathbb{1}_{LX})$ represents η_{LX}. So we see that $\beta_{X,Y}$ is given by the formula

$$k \mapsto \mathrm{Hom}(X, R(k))(u_X) = R(k)u_X. \tag{2.22}$$

The fact that this map is a bijection, implies that (LX, u_X) satisfies the following statement: Let $g \in \mathrm{Hom}_{\mathcal{B}}(X, RY)$, then there exists a unique morphism $\tilde{g} \in \mathrm{Hom}_{\mathcal{A}}(LX, Y)$ such that the diagram below commutes.

$$\begin{array}{ccc} X & \xrightarrow{u_X} & RLX \\ {\scriptstyle g}\downarrow & \swarrow_{R(\tilde{g})} & \\ RY & & \end{array}$$

So we conclude that (LX, u_X) is a universal from X to R. Now, for every $Y \in \mathcal{A}$, define $v_Y : LRY \to Y$ as the unique morphism making the diagram below commutes.

$$\begin{array}{ccc} RY & \xrightarrow{u_{RY}} & RLRY \\ {\scriptstyle \mathbb{1}_{RY}}\downarrow & \swarrow_{R(v_Y)} & \\ RY & & \end{array}$$

We claim that the pair (RY, v_Y) is a universal from L to Y, i.e., for every $f \in \mathrm{Hom}_{\mathcal{A}}(LX, Y)$ there exists a unique morphism $\tilde{f} \in \mathrm{Hom}_{\mathcal{B}}(X, RY)$ such that

$$\begin{array}{ccc} Y & \xleftarrow{v_Y} & LRY \\ {\scriptstyle f}\uparrow & \nearrow_{L(\tilde{f})} & \\ LX & & \end{array}$$

This is equivalent to prove that $\xi_{X,Y} : \mathrm{Hom}_{\mathcal{B}}(Y, RX) \to \mathrm{Hom}_{\mathcal{A}}(LY, X) : f \mapsto v_Y L(f)$ is a bijection. In fact we have

$$\begin{aligned} \beta_{X,Y}\xi_{X,Y}(f) &= R(v_Y L(f))u_X \\ &= R(v_Y)RL(f)u_X \\ &= R(v_Y)u_{RX}f \\ &= \mathbb{1}_{RY}f \\ &= f. \end{aligned} \tag{2.23}$$

Then we see that $\xi_{X,Y} = \beta_{X,Y}^{-1}$. □

Problem 5. Deduce the classical version of Hilbert 90 from the more general cohomological statement, using the "convenient complex" for $H^\bullet(\mathbb{Z}/n, -)$ that we met in class:

Theorem 0.2. Let L/K be a Galois extension, and suppose that

$$G = \mathrm{Gal}(L/K)$$

is cyclic,

$$G = \langle \sigma \rangle \cong \mathbb{Z}/n.$$

Then for every $\alpha \in L$ of norm $N(\alpha) = 1$, there exists $\beta \in L$ such that

$$\alpha = \frac{\sigma(\beta)}{\beta}.$$

Solution: The cohomological statement of Hilbert 90's is $H^1(G, L^\times) = 0$, where $G = \mathrm{Gal}(L|K)$ is the Galois group of the Galois extension $L|K$. Take the periodic projective resolution of $\mathbb{Z}$.

$$\mathbb{Z}G \xrightarrow{N} \mathbb{Z}G \xrightarrow{\sigma-1} \mathbb{Z}G \xrightarrow{\epsilon} \mathbb{Z} \to 0. \tag{2.24}$$

Now, apply the left exact contravariant functor $\mathrm{Hom}(-, L^\times)$ to the cochain complex:

$$0 \to \mathrm{Hom}_{\mathbb{Z}G}(\mathbb{Z}G, L^\times) \xrightarrow{-\circ(\sigma-1)} \mathrm{Hom}_{\mathbb{Z}G}(\mathbb{Z}G, L^\times) \xrightarrow{-\circ N} \cdots \tag{2.25}$$

Now we have an isomorphism $\mathrm{Hom}_{\mathbb{Z}G}(\mathbb{Z}G, L^\times) \to L^\times : f \mapsto f(1)$, with inverse given by $a \mapsto [f_a : g \mapsto g(a)]$. Moreover, we see that $f_a \circ (\sigma - 1)(1) = f_a(\sigma - 1) = (\sigma - 1)f_a(1) = (\sigma - 1)a = \sigma(a)a^{-1}$ and $f_a \circ N(1) = f_a(N) = Nf_a(1) = a\sigma(a) \cdots \sigma^{n-1}(a) = N_{L|K}(a)$. So, the statement $H^1(G, L^\times) = 0$, means that $\mathrm{Im}(\sigma - 1) = \mathrm{Ker}(N_{L|K})$. Therefore, if $N_{K|L}(a) = 1$, then there exists $\beta \in L^\times$ such that $\alpha = (\sigma - 1)\beta = \sigma(\beta)\beta^{-1}$. □

Problem 6. Let G be a group and $N \subset G$ a normal subgroup. If A is a G-module, verify that the invariants A^N carries a natural action of the quotient group G/N. Thus we may consider the functor

$$(-)^N : G\text{-mod} \longrightarrow (G/N)\text{-mod}.$$

Solution: The action $gN \cdot a = ga$, defines $G/N-$module structure on A^N. It's well defined because $gN = g'N \Rightarrow g^{-1}g' = n \in N$. Therefore $g^{-1}g'a = a \Rightarrow g'a = ga$. Furthermore $ga \in A^N$, $\forall g \in G$ because if $n \in N$, then there exists $n' \in N$ such that $ng = gn'$, so $nga = gn'a = ga$. □

Problem 7. Let $F : \mathcal{A} \to \mathcal{B}$ be a left exact functor, and $E : \mathcal{B} \to \mathcal{C}$ an exact functor. Assume $\mathcal{A}$ has enough injectives. Prove that

$$[R^n(E \circ F)](A) \cong [E \circ R^n F](A)$$

for all $A \in \mathcal{A}$. Deduce that there exists a natural action of G/N on the cohomology

$$H^\bullet(N, A)$$

(where G, N, A as in Exercise 6).

TODO: finnish this solution

Solution: Let $I^\bullet = (0 \to I^1 \to I^2 \to \cdots)$ be a injective resolution of $A \in \mathcal{A}$. Then

$$\begin{aligned}
[R^n(E \circ F)](A) &= H^n(0 \to E \circ F(I^1) \to E \circ F(I^2) \to \cdots) \\
&= \mathrm{Coker}(\gamma^n : \mathrm{Im}(E \circ F(d^{n-1})) \to \mathrm{Ker}(E \circ F(d^n))) \\
&= E(\mathrm{Coker}(\gamma^n : \mathrm{Im}(F(d^{n-1})) \to \mathrm{Ker}(F(d^n)))) \\
&= [E \circ R^n F](A).
\end{aligned} \tag{2.26}$$

The last step follows because E preserves kernels and cokernels.

Problem 8. Let G be a group and $N \subset G$ a normal subgroup, and A a G-module. Consider the bar complex, whose terms are

$$C^p(N, A) = \text{Maps}(N^p \to A).$$

Check that

$$[g \cdot \varphi](n_1, \ldots, n_p) = g\varphi(g^{-1}n_1 g, \ldots, g^{-1}n_p g),$$

for all $p \in \mathbb{Z}_+$, defines a G-action, and a G/N-action, on

$$H^\bullet(N, A).$$

Solution: Let $\mathcal{D}$ be the category of pairs (G, A), where A is a $G-$module, with morphisms given by $(\alpha, f) : (G, A) \to (G', A')$, where $\alpha : G \to G'$ and $f : A' \to A$ are group morphisms such that

$$f(\alpha(g)a) = gf(a),\ \forall g \in G, a \in A' \quad (*). \tag{2.27}$$

Now we claim that $H^k : \mathcal{D} \to \underline{\text{Ab}} : (G, A) \mapsto H^k(G, A)$ is a functor. So we need to prove that a morphism (α, f) induces a morphism $H^k(\alpha, f) : H^k(G, A) \to H^k(G', A')$. Let $P_k = \mathbb{Z}G^{k+1}, P'_k = \mathbb{Z}(G')^{k+1}$, so $P_\bullet = (P_k)_{k\geq 0}$ and $P'_\bullet = (P'_k)_{k\geq 0}$ are projective resolutions of $\mathbb{Z}$ viewed as a trivial module of G and G' respectively. Using α, we can regard both chains as chains of G-modules. However, the second one is no more projective as a G-module, but it's still acyclic. So we can lift the identity $\mathbb{Z} \to \mathbb{Z}$ to a chain morhism $\tau : P_\bullet \to P'_\bullet$, so it satisfies $\tau_k(gm) = \alpha(g)t_k(m),\ m \in P_k$ (also note that any two choice of liftings are homotopic). Now we define the map

$$\phi_k : \text{Hom}_{\mathbb{Z}G'}(P'_k, A') \to \text{Hom}_{\mathbb{Z}G}(P_k, A) : \varphi \mapsto f \circ \varphi \circ \tau_k. \tag{2.28}$$

We observe that

$$\begin{aligned} \phi_k(\varphi)(gm) &= f\varphi\tau_k(gm) \\ &= f\varphi(\alpha(g)\tau_k(m)) \\ &= f(\alpha(g)\varphi\tau_k(m)) \\ &= g\phi_k(\varphi)(m) \quad \text{(because of } (*)) \end{aligned} \tag{2.29}$$

Also, we observe that the differentials in the cochain complex $(\text{Hom}_{\mathbb{Z}G}(P_k, A))_{k\geq 0}$ are given by $\text{Hom}(d, A) = (-) \circ d$, where $d(g_0, \ldots, g_k) = \sum_i (-1)^i (g_0, \ldots, \hat{g}_i, \ldots, g_k)$. Using this expression it's easy to see that $d \circ \phi_k = \phi_k \circ d$. Therefore, ϕ_k descend to morphisms in cohomology:

$$\text{Hom}(f, \tau) : H^k(G', A') \to H^k(G, A) : [\varphi] \mapsto [f\varphi\tau_k] \quad (**). \tag{2.30}$$

We remark that if $\tau \sim \tau'$, then $\tau' - \tau = dh + hd$ and

$$f\varphi\tau' - f\varphi\tau = f\varphi dh + hdf\varphi. \tag{2.31}$$

If φ is a cocycle, then $\mathrm{Hom}(d, A)\varphi = \varphi \circ d = 0$. So we obtain

$$f\varphi\tau' - f\varphi\tau = f\varphi hd = \mathrm{Hom}(d, A)(f\varphi h) \tag{2.32}$$

and $[f\varphi\tau] = [f\varphi\tau']$.

Now, let $g \in G$ and $N \subset G$ a normal subgroup. Then we have a morphism $(\alpha, f) : (N, A) \to (N, A)$ given by $\alpha : N \to N : n \mapsto gng^{-1}$ and $f(a) = g^{-1}a$. The axiom $(*)$ is thus verified:

$$f(\alpha(n)a) = f(gng^{-1}a) = ng^{-1}(a) = nf(a). \tag{2.33}$$

Then we have an induced G action on $H^k(N, A)$. We denote this action by $g \mapsto c(g)^*$. This action is the same as the one stated in the problem because we have the lift τ given by $\tau_k : (n_0, \ldots, n_k) \mapsto (\alpha(n_0), \ldots, \alpha(n_k))$, so the induced action on $H^k(G, A)$ is $c(g)^* : [\varphi] \mapsto [(n_0, \ldots, n_k) \mapsto g^{-1}\varphi(gn_0g^{-1}, \ldots, gn_kg^{-1})]$. If $g \in N$, we can take the lift τ' defined by $\tau'_k(m) = gm$. It satisfies $\tau'_k(nm) = gnm = gng^{-1}gm = \alpha(n)\tau_k(m)$, $\forall n \in N$. Therefore, the associated action on $H^k(N, A)$ is given by $c(g)^* : [\varphi] \mapsto [m \mapsto g^{-1}\varphi(gm)]$, which is the identity because φ is a N-module morphism. So, if $g \in N$ we have $\tau \sim \tau'$ and $c(g)^*$ is the identity morphism. □

Problem 9. We continue in the context of the previous exercise. But now G is the symmetric group S_3 and $N \cong \mathbb{Z}/3$ is the normal subgroup generated by a 3-cycle. Consider $\mathbb{Z}$ equipped with the trivial G-module structure. Determine the G/N-module structure of the

$$H^k(N, \mathbb{Z}).$$

TODO: finnish this solution

Solution: Write $C^k(N, A) = \mathrm{Hom}_{\mathbb{Z}N}(P_k, \mathbb{Z})$. We have a basis of P_k given by tuples $(1, g_1, \ldots, g_k)$, $g_i \in N$. Let $N = \langle x : x^3 = 1\rangle$, then we use the notation

$$x_1^{\alpha_1} \cdots x_k^{\alpha_k} = (1, x^{\alpha_1}, \ldots, x^{\alpha_k}) \tag{2.34}$$

so we think the expoenents as integers modulo 3. Since $\mathbb{Z}$ is the trivial module, we can realize a morphism $\varphi \in C^k(N, A)$ as a formal sum

$$\sum_{\alpha_i \in \{0,1,2\}} a_\alpha x_1^{\alpha_1} \cdots x_k^{\alpha_k},\ \alpha_i \in \mathbb{Z}. \tag{2.35}$$

Now, let $y = (12)$, then $yxy = x^2$. We use the action induced by the lift τ in the solution of the previous problem. So

$$y \cdot \Big(\sum_{\alpha_i \in \{0,1,2\}} a_\alpha x_1^{\alpha_1} \cdots x_k^{\alpha_k}\Big) = \sum_{\alpha_i \in \{0,1,2\}} a_\alpha x_1^{2\alpha_1} \cdots x_k^{2\alpha_k}. \tag{2.36}$$

while x acts trivialy as it must be. Since $G/N = \{N, yN\}$, this completely determines the action on the set $C^k(N, A)$.

2.3 List 3

Problem 1. Write out the complex $C^{\text{ori}}(\Delta^2)$ explicitly, as well as its augmented version, and write out the homotopy h (from class) from Id to 0 (which shows that $H_n(C^{\text{ori}}(\Delta^2))$ is $\mathbb{Z}$ in degree 0 and 0 in all other degrees).

Solution: We have the following chains

$$\begin{aligned} C_0^{\text{ori}}(\Delta^2) &= \mathbb{Z}a + \mathbb{Z}b + \mathbb{Z}c \\ C_1^{\text{ori}}(\Delta^2) &= \mathbb{Z}\langle ab\rangle + \mathbb{Z}\langle ac\rangle + \mathbb{Z}\langle bc\rangle \\ C_2^{\text{ori}}(\Delta^2) &= \mathbb{Z}\langle abc\rangle \\ C_{\geq 3}^{\text{ori}}(\Delta^2) &= 0 \end{aligned} \tag{2.37}$$

Now we have complexes (we will write C_i for simplicity)

$$\begin{aligned} C &= (0 \to C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \to 0) \\ \widetilde{C} &= (0 \to C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\epsilon} \mathbb{Z} \to 0) \end{aligned} \tag{2.38}$$

where $\epsilon : C_0 \to \mathbb{Z} : a \mapsto 1, b \mapsto 1, c \mapsto 1$ and

$$\begin{aligned} \partial_1\langle ab\rangle &= b - a \\ \partial_1\langle ac\rangle &= c - a \\ \partial_1\langle bc\rangle &= c - b \\ \partial_2\langle abc\rangle &= \langle bc\rangle - \langle ac\rangle + \langle ab\rangle \end{aligned} \tag{2.39}$$

Now, we want to find a homotopy from the identity to 0 from the augmented complex to itself.

$$\begin{array}{ccccccccccc} 0 & \longrightarrow & C_2 & \longrightarrow & C_1 & \longrightarrow & C_0 & \longrightarrow & \mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow & \swarrow^{h_1} & \downarrow & \swarrow^{h_0} & \downarrow & \swarrow^{h_{-1}} & \downarrow & & \\ 0 & \longrightarrow & C_2 & \longrightarrow & C_1 & \longrightarrow & C_0 & \longrightarrow & \mathbb{Z} & \longrightarrow & 0 \end{array}$$

For the first two blocks of the diagram below we (must) have

$$\begin{aligned} &\text{(a) } \epsilon h_{-1} = \mathrm{Id}_0 \\ &\text{(b) } h_{-1}\epsilon + \partial_1 h_0 = \mathrm{Id}_{C_0} \\ &\text{(c) } h_{n-1}\partial_n + \partial_{n+1}h_n = \mathrm{Id}_{C_n} \end{aligned} \tag{2.40}$$

In particular we have $\epsilon h_{-1}(1) = 1$, so if we put $h_{-1}(1) = a$ this equation is satisfied. Now, we will use equation (b) and (c) to define h_n inductively. Using equation (b) we obtain

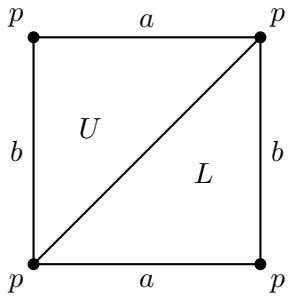
$$\begin{aligned} \partial_1 h_0(a) &= 0 \\ \partial_1 h_0(b) &= b - a \\ \partial_1 h_0(c) &= c - a \end{aligned} \tag{2.41}$$

By equations (2.39), we can put $h_0(x) = \langle ax \rangle,\ x = a, b, c$. In a similar way, by applying (c) to $\langle bc \rangle$ (for $n = 1$), we get

$$\partial_2 h_1 \langle bc \rangle + h_0(c - b) = \langle bc \rangle \Rightarrow \partial_2 h_1 \langle bc \rangle = \langle bc \rangle - \langle ac \rangle + \langle ab \rangle. \tag{2.42}$$

This suggests the general rule $h\langle w_0 w_1 \cdots w_n \rangle = \langle a w_0 \cdots w_n \rangle$. This is the homotopy between the identity map and 0, which proves that $H_n(\Delta^2) = 0$. □

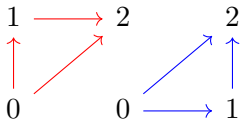
Problem 2. The following picture represents a triangulation of a 2-dimensional torus T. More precisely, the geometric realisation of the Δ-set S depicted below, is a 2-dimensional torus T.



(a) Describe the Δ-set S explicitly;

(b) Compute the homology of $C_\bullet(S)$.

Solution: First we order the vertices acording to the figure below (for the vertices of triangles U and L)



Then we define $S_0 = \{p\}, S_1 = \{a, c, b\}, S_2 = \{U, L\}$ and $S_{\geq 3} = \varnothing$ with morphisms

$$\partial_0, \partial_1 : S_1 \to S_0 : \begin{cases} a \mapsto p \\ b \mapsto p \\ c \mapsto p \end{cases} \text{ and } S_2 \to S_1 : \begin{cases} U \overset{\partial_0}{\mapsto} a,\ U \overset{\partial_1}{\mapsto} c,\ U \overset{\partial_2}{\mapsto} b \\ L \overset{\partial_0}{\mapsto} b,\ L \overset{\partial_1}{\mapsto} c,\ L \overset{\partial_2}{\mapsto} a \end{cases} \tag{2.43}$$

Now we have the chain complex

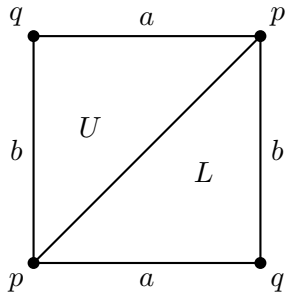
$$C_\bullet = (0 \to C_2 \overset{d_2}{\to} C_1 \overset{d_1}{\to} C_0 \to 0) \tag{2.44}$$

where $C_0 = \mathbb{Z}p, C_1 = \mathbb{Z}a + \mathbb{Z}b + \mathbb{Z}c$ and $C_2 = \mathbb{Z}U + \mathbb{Z}L$. But $d_1(a) = p - p = 0$, etc..., so $d_1 = 0$. Moreover $d_2U = a - c + b, d_2L = b - c + a$, so d_2 has matrix representation

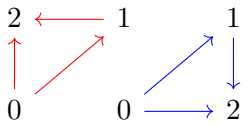
$$[d_2] = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \end{pmatrix} \overset{\text{SNF}}{\sim} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}. \tag{2.45}$$

Therefore $\text{Ker}d_2 \cong \mathbb{Z}^{\#\text{collumns}-\text{rank}(d_2)} = \mathbb{Z}^1$ and $\text{Im}(d_2) \cong \mathbb{Z}^{\text{rank}(d_2)} = \mathbb{Z}^1$. So we have $H_0(S) = \mathbb{Z},\ H_1(S) = \mathbb{Z}^3/\mathbb{Z} \cong \mathbb{Z}^2$ and $H_1(S) \cong \mathbb{Z}$. □

Problem 3. Compute the homology $H_\bullet(X, \mathbb{Z})$ and cohomology $H^\bullet(X, \mathbb{Z})$ of the real projective plane $\mathbb{R}P^2$ using the following triangulation:



Solution: First we order the vertices acording to the figure below (for the vertices of triangles U and L)



Then we have the $\Delta-$set $S_0 = \{p, q\}, S_1 = \{a, b, c\}$ and $S_2 = \{U, L\}$, with morphisms

$$\begin{aligned} &\partial_0 a = q\ \partial_1 a = p \\ &\partial_0 b = q,\ \partial_1 b = p \\ &\partial_0 c = p\ \partial_1 c = p \\ &\partial_0 U = a,\ \partial_1 U = b,\ \partial_2 U = c \\ &\partial_0 L = b,\ \partial_1 L = a,\ \partial_2 L = c \end{aligned} \tag{2.46}$$

So, the differentials are

$$\begin{aligned} &d_1 a = d_1 b = q - p,\ d_1 c = 0 \\ &d_2 U = a - b + c,\ d_2 L = b - a + c \end{aligned} \tag{2.47}$$

Then we have matrix representations

$$[d_2] = \begin{pmatrix} 1 & -1 \\ -1 & 1 \\ 1 & 1 \end{pmatrix} \overset{\text{SNF}}{\sim} \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} \text{ and } [d_1] = \begin{pmatrix} 1 & 1 & 0 \\ -1 & -1 & 0 \end{pmatrix} \overset{\text{SNF}}{\sim} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{2.48}$$

Since d_2 has maximal rank, it's injective and $H_2(\mathbb{R}P^2) = (0)$. Furthermore, by the smith normal form of the matrix representation of d_2, we get $\text{Im}(d_2) \cong \mathbb{Z} \oplus 2\mathbb{Z}$. Also, we have $\text{Ker}(d_1) \cong \mathbb{Z}^{3-1} = \mathbb{Z}^2$, so $H_1(\mathbb{R}P^2) = \mathbb{Z}^2/(\mathbb{Z} \oplus 2\mathbb{Z}) \cong \mathbb{Z}/(2)$. Finally, we have $H_0(\mathbb{R}P^2) \cong \mathbb{Z}^2/\mathbb{Z} \cong \mathbb{Z}$.

For cohomology, we have the co-complex $C^n = \text{Hom}(C_n, \mathbb{Z})$ with differentials given by $d^n = \text{Hom}(d_{n+1}, \mathbb{Z})$. In particular, matrix representations are obtained by transposing the previous ones.

$$[d^0] = [d_1]^T = \begin{pmatrix} 1 & -1 \\ 1 & -1 \\ 0 & 0 \end{pmatrix} \overset{\text{SNF}}{\sim} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \; [d^1] = [d_2]^T = \begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix} \overset{\text{SNF}}{\sim} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} \tag{2.49}$$

So we obtain

$$H^0(\mathbb{R}P^2) = \mathbb{Z}, \; H^1(\mathbb{R}P^2) = (0) \text{ and } H^2(\mathbb{R}P^2) = \mathbb{Z}/(2). \tag{2.50}$$

□

Problem 4. Check that the universal coefficient theorem for cohomology confirms the result of Question 3.

Solution: The universal theorem for cohomology says

$$H^n(X, R) \cong \text{Hom}_R(H_n(X, R), R) \oplus \text{Ext}^1_R(H_{n-1}(X, R), R). \tag{2.51}$$

Then we have

Degree 0:

$$\text{Hom}_{\mathbb{Z}}(H_0(X), \mathbb{Z}) \oplus \text{Ext}^1_{\mathbb{Z}}((0), \mathbb{Z}) = \text{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z} \tag{2.52}$$

Degree 1:

$$\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/(2), \mathbb{Z}) \oplus \text{Ext}^1(\mathbb{Z}, \mathbb{Z}) = (0) \tag{2.53}$$

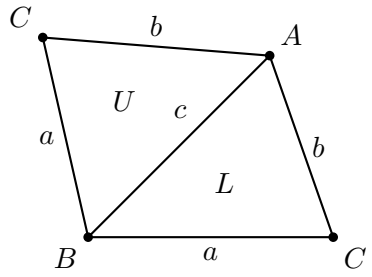
Degree 2:

$$\text{Hom}_{\mathbb{Z}}((0), \mathbb{Z}) \oplus \text{Ext}^1(\mathbb{Z}/(2), \mathbb{Z}) = \mathbb{Z}/(2) \tag{2.54}$$

which agrees with the cohomologies calculated in the previous problem. □

Problem 5. Compute the homology of the n-sphere S^n. [Hint: You might want to present S^n as a pair of n-simplices glued along their boundary. The result of Question 1 above might help.]

Solution: First, we will do the calculation for $n = 2$ using the triangulation suggested.



Then we have a chain given by $C_0 = \mathbb{Z}A+\mathbb{Z}B+\mathbb{Z}C, C_1 = \mathbb{Z}a+\mathbb{Z}b+\mathbb{Z}c, C_2 = \mathbb{Z}U+\mathbb{Z}L,\ C_{\geq 3} = 0$ with differentials

$$d : C_2 \to C_1 : \begin{cases} dU = a - b + c \\ dL = a - b + c \end{cases} \text{ and } d : C_1 \to C_0 : \begin{cases} da = C - B \\ db = C - A \\ dc = B - A \end{cases} \tag{2.55}$$

Therefore, we have the following matrix representations

$$[d_2] = \begin{pmatrix} 1 & 1 \\ -1 & -1 \\ 1 & 1 \end{pmatrix} \overset{\text{SNF}}{\sim} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix},\ [d_1] = \begin{pmatrix} 0 & -1 & -1 \\ -1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \overset{\text{SNF}}{\sim} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{2.56}$$

Then we obtain

$$H_n(\mathbb{S}^2) = \begin{cases} \dfrac{\mathbb{Z}}{(0)} \cong \mathbb{Z}, n = 2 \\ \dfrac{\mathbb{Z}}{\mathbb{Z}} = (0),\ n = 1 \\ \dfrac{\mathbb{Z}^3}{\mathbb{Z}^2} \cong \mathbb{Z},\ n = 0 \\ 0,\ n \geq 3 \end{cases} \tag{2.57}$$

From this calculation we see a general rule: $C_k(\mathbb{S}^n) = C_k(\Delta^n)$ for $k = 0, \ldots, n-1$ because all the edges of the two complexes Δ^n are identified. In particular $H_k(\mathbb{S}^n) = H_k(\Delta^n)$ $(k = 0, \ldots, n-1)$. So, if $\Delta^n = [v_0 \ldots v_n]$, then $C_n = \mathbb{Z}U + \mathbb{Z}L$ and the differential $d : C_n \to C_{n-1}$ gives

$$dU = \sum (-1)^i [v_0 \cdots \hat{v}_i \cdots v_n] = dL. \tag{2.58}$$

So d has a one dimensional kernel given by $\mathbb{Z}(U - L)$ and $H_n(\mathbb{S}^n) = \mathbb{Z}^1/(0) = \mathbb{Z}$. □

Problem 6. Let $\mathcal{C}_\bullet$ and $\Delta_\bullet$ be two complexes (of objects of some abelian category) and let $f_\bullet, g_\bullet : \mathcal{C}_\bullet \to \mathcal{D}_\bullet$ be two morphisms. Show that a homotopy from $f_\bullet$ to $g_\bullet$ induces a dual homotopy from $g_\bullet^*$ to $f_\bullet^*$, where $f_\bullet^*, g_\bullet^* : \mathcal{D}_\bullet^* \to \mathcal{C}_\bullet^*$ are the corresponding adjoint morphisms.

Solution: We have a homotopy h between f and g:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & C_2 & \longrightarrow & C_1 & \longrightarrow & C_0 & \longrightarrow & 0 \\ & \swarrow^{h_2} & \downarrow & \swarrow^{h_1} & \downarrow & \swarrow^{h_0} & \downarrow & & \\ 0 & \longrightarrow & D_2 & \longrightarrow & D_1 & \longrightarrow & D_0 & \longrightarrow & 0 \end{array}$$

That is, we have

$$d_{n+1}h_n + h_{n-1}d_n = f_n - g_n. \tag{2.59}$$

Now, we apply the functor $\mathrm{Hom}(-, A)$ to this equation, so we obtain

$$\mathrm{Hom}(h_n, A)\mathrm{Hom}(d_{n+1}, A) + \mathrm{Hom}(d_n, A)\mathrm{Hom}(h_{n-1}, A) = \mathrm{Hom}(f_n, A) - \mathrm{Hom}(g_n, A). \tag{2.60}$$

Furthermore, we have by definition $d^n = \mathrm{Hom}(d_{n+1}, A)$ and $h^n = \mathrm{Hom}(h_{n-1}, A)$, i.e.,

$$h^{n+1}d^n + d^{n-1}h^n = \mathrm{Hom}(f_n, A) - \mathrm{Hom}(g_n, A) \tag{2.61}$$

which defines a homotopy between $\mathrm{Hom}(f_n, A)$ and $\mathrm{Hom}(g_n, A)$. □

7. Let $C^\bullet$ be a complex of finite dimensional vector spaces, and suppose that $C^n = 0$ for all but finitely many $n \in \mathbb{Z}$. Show that

$$\sum_{n\in\mathbb{Z}} (-1)^n \dim C^n = \sum_{n\in\mathbb{Z}} (-1)^n \dim H^n(C^\bullet).$$

Solution: Suppose that we have a co-cochain complex

$$C^\bullet = (0 \to C^0 \to C^1 \to C^2 \to \cdot \to C_N \to 0 \to \cdots). \tag{2.62}$$

Then we calculate

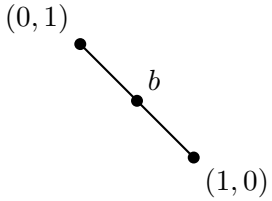
$$\begin{aligned}\sum_{n\geq 0}(-1)^n \dim H^n(C^\bullet) &= \dim \mathrm{Ker} d^0 + \sum_{n=1}^{N}(\dim \mathrm{Ker} d^n - \dim \mathrm{Im}(d^{n-1})) \\ &= \dim \mathrm{Ker} d^0 - \dim \mathrm{Ker} d^1 + \dim \mathrm{Im} d^0 \\ &\quad + \dim \mathrm{Ker} d^1 - \dim \mathrm{Im} d^1 \\ &\quad \cdots \\ &\quad (-1)^{N-1} \dim \mathrm{Ker} d^{N-1} + (-1)^N \dim \mathrm{Im} d^{N-2} \\ &\quad (-1)^N \dim \mathrm{Ker} d^N + (-1)^{N-1} \dim \mathrm{Im} d^{N-1} \\ &= \sum_{n\geq 0}(-1)^n (\dim \mathrm{Ker} d^n + \dim \mathrm{Im} d^n) \\ &= \sum_{n\geq 0}(-1)^n \dim C^n.\end{aligned} \tag{2.63}$$

□

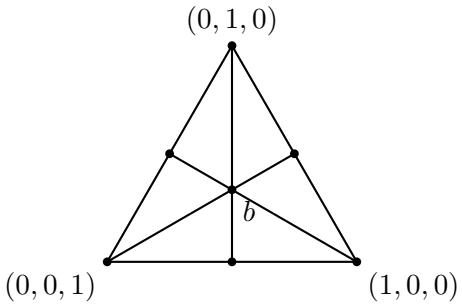
Definition 0.1. Let $\Delta^n \subset \mathbb{R}^{n+1}$ be the topological n-simplex, and $v_0, \ldots, v_n$ its vertices. Denote by $b \in \Delta^n$ the point $(\frac{1}{n+1}, \frac{1}{n+1}, \ldots, \frac{1}{n+1})$. The barycentric subdivision of Δ^n is the decomposition of into the n-simplices $[b, w_0, \ldots, w_{n-1}]$ where, inductively, $[w_0, \ldots, w_{n-1}]$ is an $(n-1)$-simplex in the barycentric subdivision of a face $[v_0, \ldots, \widehat{v_i}, \ldots, v_n]$ of Δ^n.

Problem 8. Draw the barycentric subdivisions of Δ^0, Δ^1 and Δ^2.

Solution: We have $\Delta^0 = *$, the barycentric subdivision of a point. Next, we have the barycentric subdivision of a line Δ^1.



Finally, we have the barycentric subdivision of a triangle Δ^2.



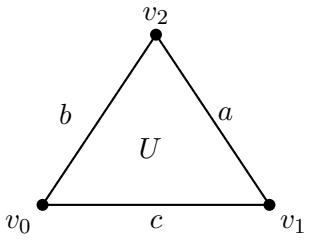
□

Problem 9. Let S be a Δ-set. We define the barycentric subdivision $\mathrm{Sd}(S)$ (which is another Δ-set) of S as follows:

$$\mathrm{Sd}(S)_n = \{(\sigma, f_1, \ldots, f_n), \sigma \in S_N, f_i : [k_i] \hookrightarrow [k_{i-1}], 0 \leq k_n < \cdots < k_0 = N\}$$

Explain the correspondence between this definition and the geometric one, on the specific example of the subdivision of Δ^2.

Solution: Take a 2-simplex Δ^2.



This is a $\Delta-$set given by $S_0 = \{v_0, v_1, v_2\}, S_1 = \{a, b, c\}, S_2 = \{U\}$ and $S_{\geq 3} = \varnothing$. Then, by definition we have

$$\mathrm{Sd}(S)_2 = \{(U, f_1, f_2) : [0] \overset{f_1}{\hookrightarrow} [1] \overset{f_2}{\hookrightarrow} [2]\}. \tag{2.64}$$

So we have $\#\mathrm{Sd}(S)_2 = 6$, which is the number of small triangles in the barycentric subdivision of Δ^2 in the problem 8. The idea is that each pair of injections f_1, f_2 parametrize a small triangle in that subdivision. In fact, each pair give a sequence of vertices os a small triangle:

$$v_{f_2 f_1(0)},\ \frac{v_{f_2 f_1(0)} + v_{f_2(1-f_1(0))}}{2},\ \frac{v_0 + v_1 + v_2}{3}. \tag{2.65}$$

For example, the pair $f_1(0) = 0, f_2 : 0 \mapsto 0, 1 \mapsto 1$ give the sequence of vertices $v_0, \frac{v_0+v_1}{2}$ and $\frac{v_0+v_1+v_2}{3}$.

Moreover, we have

$$\mathrm{Sd}(S)_1 = \bigcup_{i=0}^{1} \{(U, f) : [i] \overset{f}{\hookrightarrow} [2]\} \cup \bigcup_{x \in \{a,b,c\}} \{(x, f) : [0] \overset{f}{\hookrightarrow} [1]\}. \tag{2.66}$$

Here, the geometric correspondence is as follows. A pair $(U, f), f : [1] \to [2]$ corresponds to an edge in the barycentric subdivison of U by connecting the vertices $\frac{v_{f(0)}+v_{f(1)}}{2}$ and $\frac{v_0+v_1+v_2}{3}$. Also a pair $(U, f), f : [0] \to [2]$ corresponds to the edge connecting $v_{f(0)}$ and $\frac{v_0+v_1+v_2}{3}$. Finally, the a pair (x, f) take an edge and select one of the segments between it midpoint and another vertex. For example $(a, f), f : 0 \mapsto 1$ gives the edge connecting $\frac{v_1+v_2}{2}$ and v_2.

Finally, the set $\mathrm{Sd}(S)_0 = \{v_0, v_1, v_2, a, b, c, U\}$ corresponds to each one of the 7 points of the barycentric subdivision of Δ^2. □